

Language Agent Tree Search

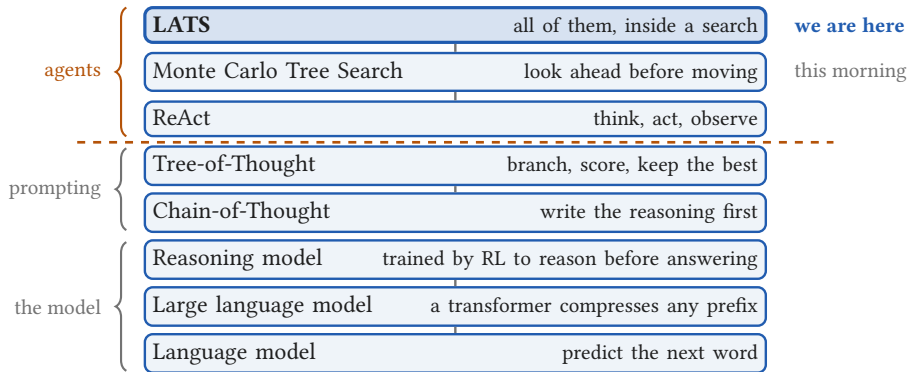


One frozen model as a planning agent

Weekend 6 · Frontier Agentic AI

Carlos Cotrini

Where LATS sits





Section 1

The LATS method

1

UCT: the critic, plus an exploration bonus

$$\text{UCT}(s) = \underbrace{V(s)}_{\text{the critic}} + \underbrace{w \sqrt{\frac{\ln N(p)}{N(s)}}}_{\text{the exploration bonus}}$$

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The critic

what the model thinks
this branch is worth

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The critic
what the model thinks
this branch is worth

+

The exploration bonus
how little we have tried it

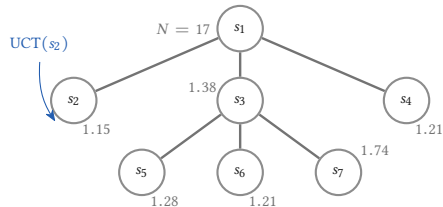
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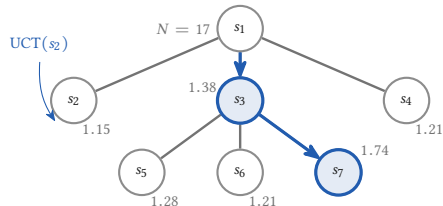


This morning's words for the same two terms: how good it looked, plus how little we know.

The same loop as this morning, with four edits



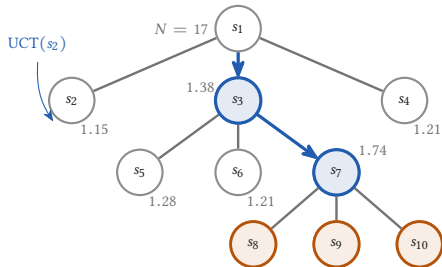
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1 Selection

by UCT

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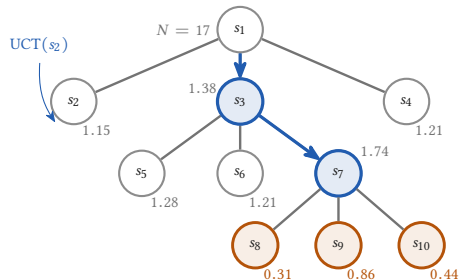
1 Selection

by UCT

2 Expansion

the *actor* proposes

The same loop as this morning, with four edits

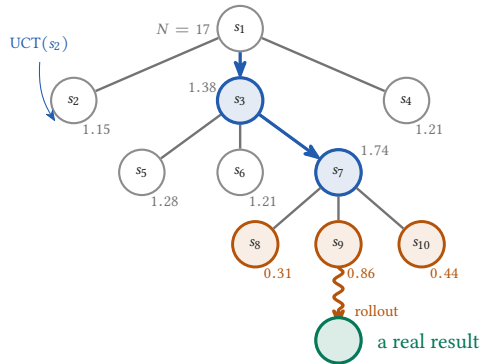


1 Selection by UCT

2 Expansion the *actor* proposes

3 Evaluation the *critic* scores

The same loop as this morning, with four edits



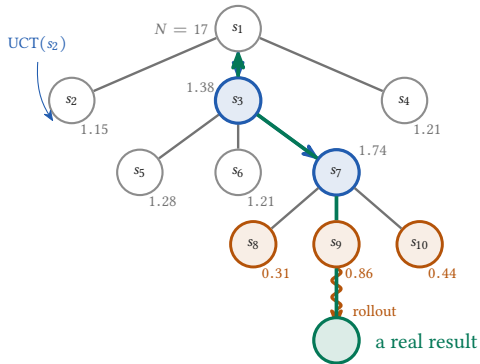
1 Selection by UCT

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4 Simulation the *best* one, to the end

The same loop as this morning, with four edits



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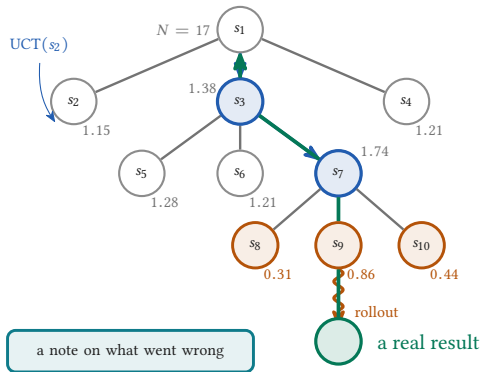
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5 Backpropagation the *real* result, up

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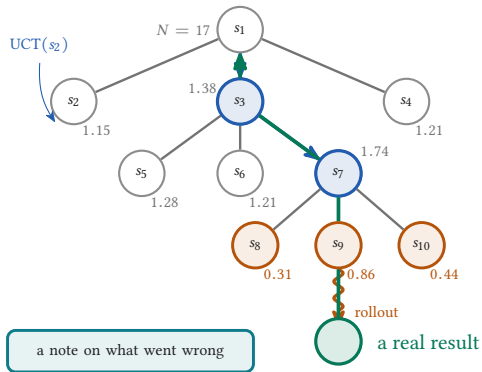
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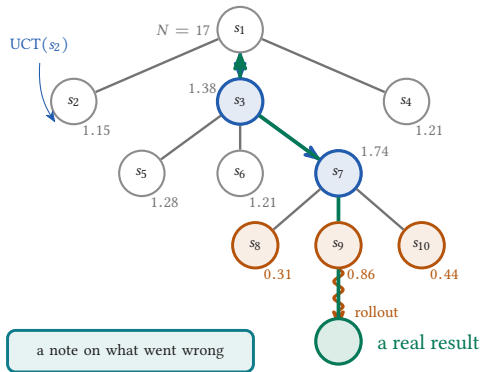
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grey: UCT, what selection compares.

amber: V from the critic, and all that is new.

The same loop as this morning, with four edits



1 Selection by UCT

2 Expansion the actor proposes

3 Evaluation the critic scores

4 Simulation the best one, to the end

5 Backpropagation the real result, up

6 Reflection the feedback model writes it

into the next prompt

grey: UCT, what selection compares.

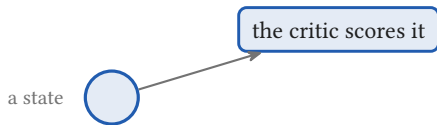
amber: V from the critic, and all that is new.

What the critic actually returns



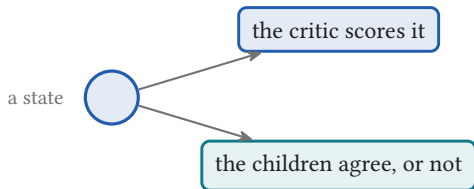
$$V(s) = \quad +$$

What the critic actually returns



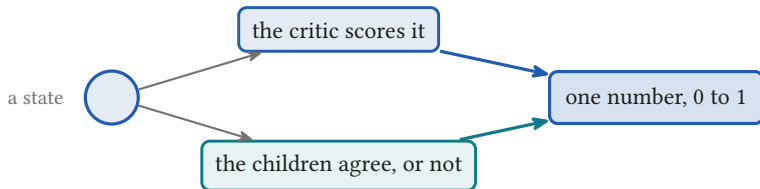
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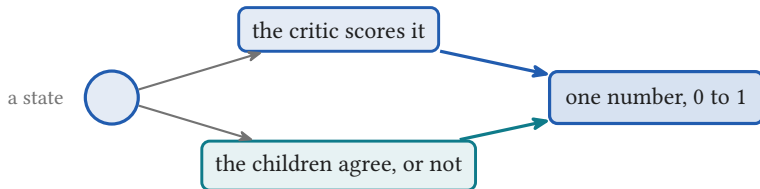
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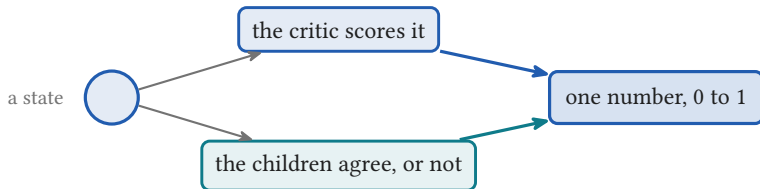
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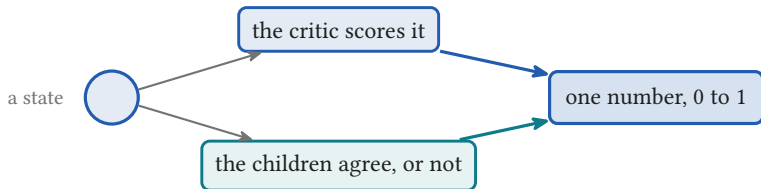
$$V(s) = \lambda LM(s) +$$

What the critic actually returns



$$V(s) = \lambda LM(s) + (1 - \lambda) SC(s)$$

What the critic actually returns



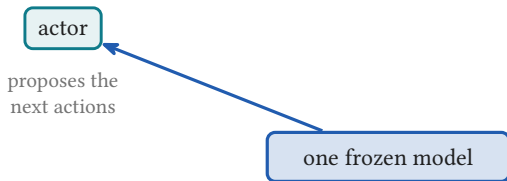
$$V(s) = \lambda LM(s) + (1 - \lambda) SC(s)$$

$\lambda = 0.5$ on question answering, 0.8 on programming and web tasks.

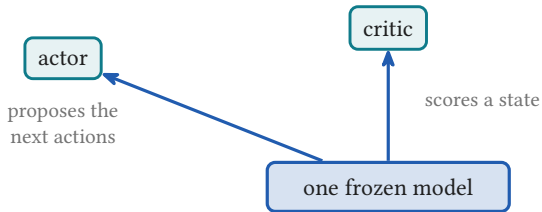
Three jobs, one set of weights

one frozen model

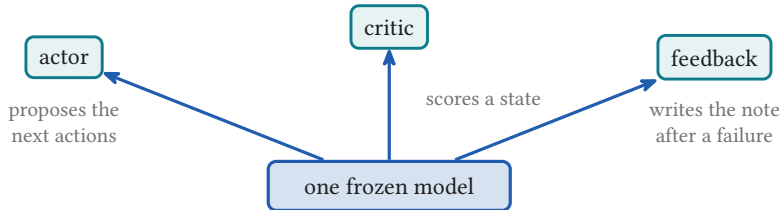
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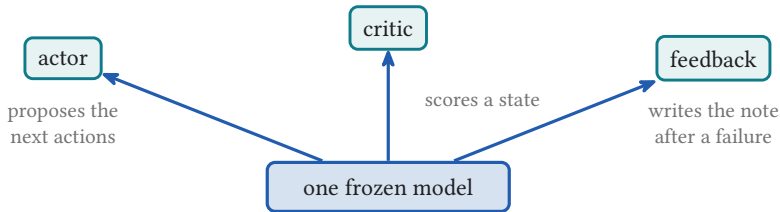
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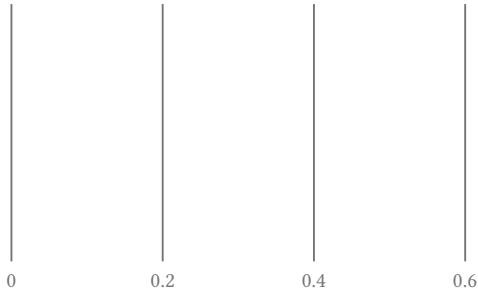


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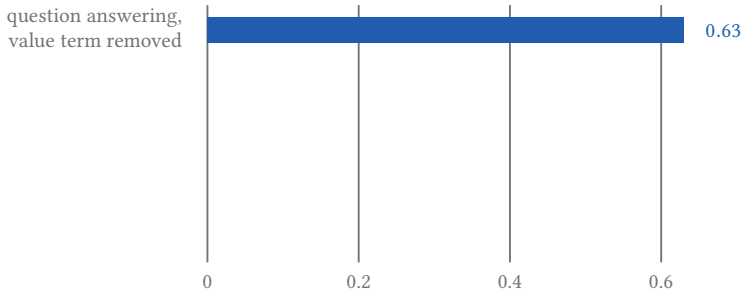


same weights every time. Only the prompt changes.

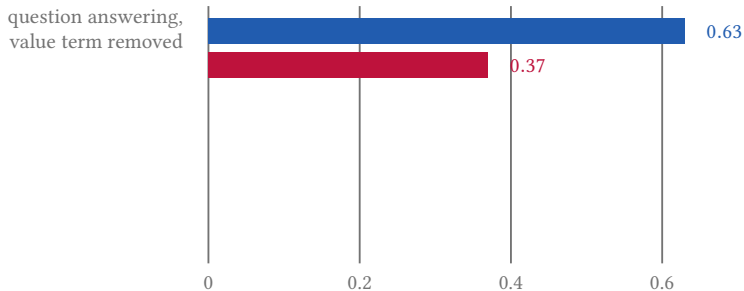
What happens when you remove each piece



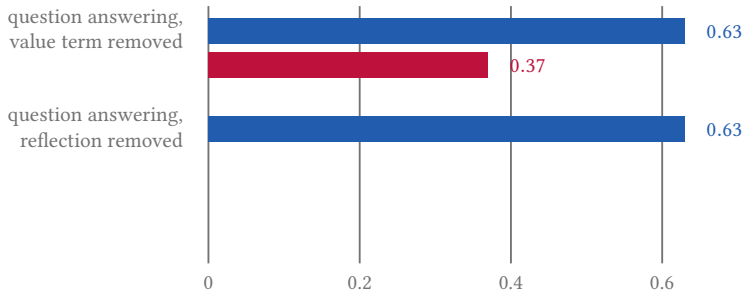
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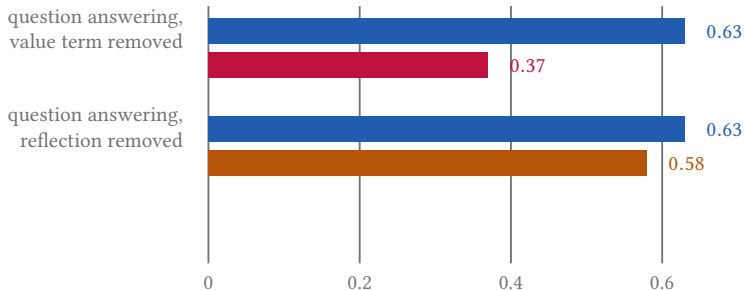
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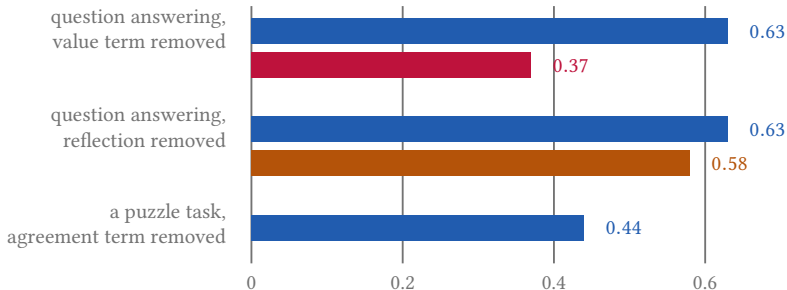
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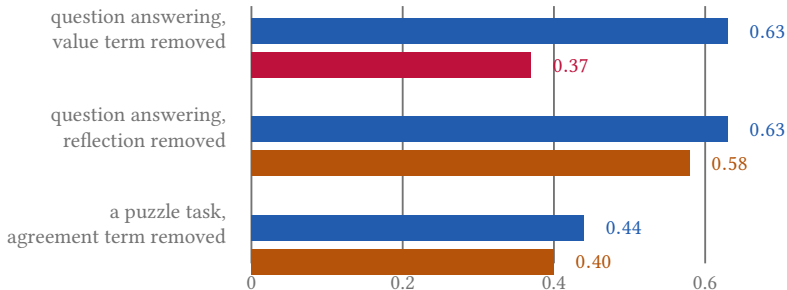
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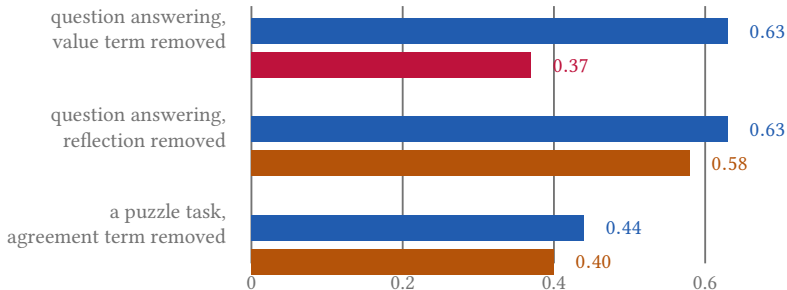
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What happens when you remove each piece



The critic's score is doing almost all of the work. The last row is a different task, so read it on its own.